

Practical-11

```
GNU nano 8.7.1 race_condition.c *
#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define ITERATIONS 100000

int counter = 0;

void *increment(void *arg)
{
    for (int i = 0; i < ITERATIONS; i++)
    {
        counter++;
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_create(&threads[i], NULL, increment, NULL);
    }

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("Expected counter value: %d\n",
        NUM_THREADS * ITERATIONS);

    printf("Actual counter value: %d\n", counter);

    return 0;
}
```

Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ nano race_condition.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ gcc race_condition.c -o race_condition -pthread
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ ./race_condition
Expected counter value: 400000
Actual counter value: 177171
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ |
```

Part:b

```
GNU nano 8.7.1 mutex_counter.c *
pthread_mutex_init(&lock, NULL);

for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_create(&threads[i], NULL, increment, NULL);
}

for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_join(threads[i], NULL);
}

printf("Expected counter value: %d\n",
    NUM_THREADS * ITERATIONS);

printf("Actual counter value: %d\n", counter);

pthread_mutex_destroy(&lock);

return 0;
}
```

Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ nano mutex_counter.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ gcc mutex_counter.c -o mutex_counter -pthread
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ ./mutex_counter
Expected counter value: 400000
Actual counter value: 400000
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ |
```

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ nano mutex_counter.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ gcc mutex_counter.c -o mutex_counter -pthread
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ ./mutex_counter
Expected counter value: 400000
Actual counter value: 400000
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ ./mutex_counter
./mutex_counter
./mutex_counter
Expected counter value: 400000
Actual counter value: 400000
Expected counter value: 400000
Actual counter value: 400000
Expected counter value: 400000
Actual counter value: 400000
Expected counter value: 400000
Actual counter value: 400000
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ ./mutex_counter
Expected counter value: 400000
Actual counter value: 400000
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ ./mutex_counter
Expected counter value: 400000
Actual counter value: 400000
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ |
```